

● Original Contribution

TUMOR CONTRAST IMAGING WITH GAS VESICLES BY CIRCUMVENTING THE RETICULOENDOTHELIAL SYSTEM

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Abstract—Gas vesicles (GVs) are nanosized structures (45–800 nm) and have been reported to produce non-linear contrast signals, making them an attractive agent for molecular targeting of tumors. One barrier to their use for pre-clinical oncology studies is rapid uptake into the reticuloendothelial system (RES) and consequent rapid decrease in contrast signal after infusion ends and low signal on reperfusion after a bubble burst sequence. The purpose of this study was to examine suppression of the RES and surface modification of GV to prolong contrast circulation in tumors for ultrasound imaging. Ultrasound imaging to measure dynamics of contrast signal intensity in tumor models was carried out using a 21-MHz high-frequency array transducer with the Vevo 2100 ultrasound system. The non-linear contrast signal from intravenously injected GV compared with peak enhancement was measured during contrast wash-out and on reperfusion after a contrast burst sequence. Disrupting the RES by saturating the macrophage population or chemically inhibiting the Kupffer cell population with gadolinium or Intralipid preserves 62%–88% of GV's contrast enhancement relative to peak during the wash-out phase and 32%–56% on reperfusion compared with 38% and 14%, respectively, for no disruption of the RES, indicating longer circulation of GV in the tumor. Additionally, coating the GV with 2-, 5- or 10-kDa polyethylene glycol (PEG) chains resulted in >70% contrast signal retention in the tumors during wash-out and, for 5- or 10-kDa PEG chains, a return to >45% of peak contrast signal on reperfusion. These findings indicate that GV can be used as a contrast agent for tumor imaging and that disruption of the RES improved recirculation and maintained contrast enhancement caused by GV in the tumors. (E-mail: c.demore@utoronto.ca) © 2019 World Federation for Ultrasound in Medicine & Biology. All rights reserved.

Key Words: Ultrasound contrast agent, Gas vesicles, Tumor imaging, Acoustic nanoparticles, Reticuloendothelial system.

INTRODUCTION

Ultrasound contrast agents (UCAs) have become a valuable tool for patient care in the medical field. Because of their high contrast-to-tissue background ratio, UCA can be used to enhance visualization of the microvasculature and assess changes in tissue perfusion, making them a very useful diagnostic tool for multiple diseases, including cancer. To date the most widely studied and commonly used UCAs are microbubbles. Microbubbles (MBs) consist of a gas core stabilized by an outer shell often made up of phospholipids, proteins or polymers (Chong et al. 2018). They range in size from 1–10 μm , approximately the same size range as a blood cell, and, as a result, are restricted to the circulatory system

(Deshpande et al. 2010; Mulvana et al. 2017). Furthermore, there has been considerable research using microbubbles as molecular UCAs (Ellegala et al. 2003; Rychak et al. 2007; Chang et al. 2017; Willmann et al. 2017). This is done by tagging the surface of microbubbles with ligands such as antibodies and small peptides to recognize receptors on targeted sites. Targeted microbubbles can be used to detect and monitor a variety of disease states. However, the endothelial lining of the vasculature is highly restrictive, allowing only nanometer-sized molecules to pass through to the interstitial space. Therefore, molecular targeting with micron-sized microbubbles are restricted to proteins expressed solely in the vasculature (Abou-Elkacem et al. 2015).

Recently, it has been reported that nanosized particles called gas vesicles are a promising new source of ultrasound contrast (Shapiro et al. 2014; Lakshmanan et al. 2016, 2017). Gas vesicles (GVs) are subcellular

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nanostructures expressed almost exclusively in aquatic micro-organisms to regulate buoyancy (Walsby 1994). Synthesized inside the bacteria, gas vesicles are gas-filled structures composed entirely of protein, predominately gas vesicle protein A (GvpA) and gas vesicle protein C (GvpC) (Walsby and Hayes 1989; Walsby 1994). Depending on the organism, the gas vesicles can vary in shape from cylindrical to lemon shape and can range from 45–200 nm or more in width and from 100–800 nm or more in length (Walsby 1994). Interestingly, unlike microbubbles, the protein shell of gas vesicles excludes water but permits the free exchange of gas with the surrounding environment (Walsby 1994; Pfeifer 2012). Shapiro et al. (2014) were the first to report that gas vesicles can produce stable ultrasound contrast signal both *in vitro* and *in vivo*. Examination of the acoustic behaviour of gas vesicles suggests that the harmonic signals produced by GVs when subjected to ultrasound are the result of buckling (Cherin et al. 2017). Ultimately, the stability of the gas core and nanometer-sized particle makes gas vesicles a very appealing agent for molecular targeting.

The GV protein nanostructures offer passive and targeted ultrasound contrast, and they offer the potential to be developed into powerful acoustic reporter genes (ARGs) (Bourdeau et al. 2018). Bourdeau et al. (2018) engineered *Escherichia coli* to express a combination of gas vesicle genes, termed ARGs, resulting in these bacteria cells producing robust ultrasound contrast signals. More interestingly, when colonized in the gut of mice, ultrasound imaging clearly revealed the localization of ARG-expressing *E. coli* cells in the colon (Bourdeau et al. 2018). (Farhadi et al. 2019). recently reported a major breakthrough in which mammalian ARGs were demonstrated to permit ultrasound imaging of gene expression in mice. This potential of GVs as ARGs or as circulating contrast agents greatly expands their value in biomedical research.

Maeda and Matsumura (1986) were among the first to speculate that the leaky nature of the tumor vasculature had an enhanced permeability and retention effect. Leaky vasculature with openings ranging from 200 nm to 1 μ m have been reported in subcutaneously implanted xenograft tumors (Yuan et al. 1995; Hobbs et al. 1998). Because of their size, the current molecular targeting acoustic contrast agents (microbubbles) are restricted to the vascular space, whereas the nanosized structure of gas vesicles offers the possibility of extravasating out of the vasculature of tumors. This makes GVs a unique agent for molecular imaging of cell surface biomarkers outside of the vasculature.

A prolonged retention of GVs in the blood, for a longer circulation time, is critical in increasing the probability that GVs can extravasate and encounter targets of interest. However, the reticuloendothelial system (RES), also known as the mononuclear phagocytic system, consists of tissue-specific macrophages that are responsible

for clearing out foreign particles (Davies et al. 2013). In a prior investigation of *in vivo* contrast imaging of intravenously injected GVs (Shapiro et al. 2014), robust contrast scattering was observed in the inferior vena cava, with the subsequent accumulation of GVs in the liver reaching a steady state. After a high-power burst to collapse the GVs, the contrast signal was eliminated. Interestingly, in the absence of a high-power burst, GVs appear to accumulate in the liver with an observed 50% loss of contrast signal after approximately 15 min, before finally being degraded within 60 min. The prolonged accumulation of GV contrast signal observed in the liver combined with the lack of reperfusion after a high-power burst in the liver, as noted by Shapiro et al. and as observed in tumor models described in this article, suggests a static accumulation of GVs in the liver rather than free-flowing GVs in the circulation. As Kupffer cells are the tissue macrophages that reside in the liver and are responsible for clearance of nanoparticles, it is highly likely the GVs are being rapidly taken up and cleared from circulation by Kupffer cells. When injected subcutaneously under the skin, GVs produced stable contrast for at least 2 h because of the lack of rapid nanoparticle clearance and protein degradation seen in the liver (Shapiro et al. 2014). Additionally, technetium-labelled GVs accumulated predominately in organs containing resident tissue macrophages such as the liver, lung and spleen (Le Floch et al. 2018). This biodistribution study further validates our theory that the RES must play a role in the uptake of GVs.

In the context of imaging pre-clinical models of disease, we evaluate the dynamics of gas vesicles as non-linear UCAs for tumour imaging. Here, we report the circumvention of the RES through saturation, chemical disruption and PEGylation to improve the preservation of the contrast signal in tumours from circulating GVs during wash-out and reperfusion after a burst sequence to collapse the acoustic protein nanostructures.

METHODS

Cell lines and reagents

Lewis lung carcinoma cells were purchased from American Type Culture Collection and cultured in Dulbecco's modified Eagle's medium (DMEM, Gibco, Life Technologies, Carlsbad, CA, USA) supplemented with 5% fetal bovine serum (Gibco Life Technologies). Gadolinium chloride (GdCl_3) was purchased from Sigma-Aldrich (St. Louis, MO, USA), and 20% Intralipid was purchased from Baxter (Mississauga, ON, Canada). Vevo MicroMarker (FUJIFILM VisualSonics, Toronto, ON, Canada) microbubbles were prepared and used per the manufacturer's instructions.

Gas vesicle preparation

GVs were isolated from *Halobacterium* NRC-1 (*Halo*) as previously described (Shapiro *et al.* 2014; Lakshmanan *et al.* 2017). Briefly, *Halobacterium* sp. NRC-1 (Carolina Biological Supply, Burlington, NC, USA) was grown in Carolina medium at 42°C for 1 wk and subsequently transferred to separatory funnels for another week to allow for buoyant *Halos* to float to the surface. Buoyant *Halos* were lysed using hypo-osmotic shock (TMC lysis buffer: 10 mM Tris, pH 7.5, 2.5 mM MgCl₂, 2 mM CaCl₂) and subjected to multiple rounds of centrifugation until the supernatant was clear and there was no evidence of remaining cell debris. Purified GVVs were resuspended in phosphate-buffered saline (PBS), and concentrations were determined by measuring the optical density at 500 nm (OD₅₀₀) using a Nanodrop 2000 (ThermoFisher Scientific, Waltham, MA, USA) (Shapiro *et al.* 2014). The number of GVVs per millilitre (GVVs/mL) was calculated based on the molar concentration of 20 pM/OD₅₀₀ as determined by Shapiro *et al.* (2014).

PEGylation of GVVs

Functionalization of GVVs with polyethylene glycol (PEG) was achieved with the labeling method utilizing the succinimidyl carboxymethyl ester labeling reagent. Methoxy-PEG (2, 5, 10 and 15 kDa) succinimidyl carboxymethyl ester was purchased from Jenkem Technology (Plano, TX, USA). The succinimidyl carboxymethyl ester group reacts with the amino group on proteins (–NH₂) in pH 7–9 buffers to form a stable amide bond, thus covalently linking the PEG chain on lysine residues to the protein shell of GVVs. Succinimidyl carboxymethyl ester PEG chains were reacted with gas vesicles in PBS, pH 7.4, at 100 × molar excess of PEG-esters to gas vesicles for 45 min at room temperature with gentle rocking. Multiple rounds of centrifugation were used to remove excess unbound PEG-esters and followed by resuspension in PBS. The concentrations of PEGylated GVVs were quantified as described above by measuring the optical density. Confirmation of successful PEGylation of GVVs was carried out using a commercially available mPEG ELISA kit (Life Diagnostics, West Chester, PA, USA) according to the manufacturer's instructions.

Formation of xenografts in SHO mice

Lewis Lung Carcinoma (LLC) cells were individualized, and 5 × 10⁵ cells in 50 µL DMEM were injected subcutaneously (s.c.) into the hindlimbs of 5- to 7-wk-old female SHO mice (Charles River Laboratory, Wilmington, MA, USA). Tumors were left to grow for 12–14 d and imaged when tumors reached a depth of 5–7 mm. All experimental procedures were approved by the Animal Care Committee at Sunnybrook Research Institute.

Ultrasound imaging and analysis

Ultrasound tumour imaging was performed with a Vevo 2100 (FUJIFILM VisualSonics) scanner and 21-MHz high-frequency linear array transducer (MS250, FUJIFILM VisualSonics) using the non-linear contrast mode with the following parameters: frequency 18 MHz, transmit power 2%, contrast gain 32 dB, beamwidth wide, frame rate 15, 6- to 10-mm foci, dynamic range 35 dB, axial and lateral resolution of 75 and 165 µm, respectively. Animals were anesthetized using isoflurane (4% initiation, 2% maintenance) during the imaging session. GVVs and microbubbles were injected through a tail vein catheter at the rate of 500 µL/min using an injection pump. To compare the wash-in and wash-out kinetics of MBs GVVs in tumours, two animals were injected intravenously (i.v.) with a 50-µL bolus of microbubbles at 2 × 10⁹ MBs/mL, followed 20 min later, (as stated by the manufacturer) to allow for the clearance of MBs from circulation, by an injection of native GVVs. For all gas vesicle injections, 300 µL of native GVVs (n = 21) or functionalized GVVs (2 kDa n = 5, 5 kDa n = 7, 10 kDa n = 8, 15 kDa n = 5) were injected i.v. at OD₅₀₀ 40 (4.8 × 10¹¹ GVVs/mL). The first and simplest strategy tested to overcome the RES was to saturate the liver and spleen with an initial injection of 300 µL (OD 40) of native gas vesicles and then perform a second injection with the same volume and OD after 20 min (n = 4). To examine the effects of Intralipid and GdCl₃ on RES inhibition, animals were first injected with native GVVs as control, followed by a high-power (100% transmit power) 3-D burst sequence applied to the entire liver to collapse residual GV particles. Mice were allowed to rest for 1–2 h before treatment with GdCl₃ (n = 6, 10 mg/kg) or Intralipid (n = 5, 4 g/kg) followed by a second round of imaging with native GVVs at 24–36 or 1 h, respectively. PEGylated GVVs were examined in mice that had not been treated with any RES inhibition nor given a prior injection of GVVs.

The imaging sequence for both MBs and GVVs included a 15-s baseline acquisition, 36-s infusion of GVVs and, at 110 s, the application of a 1-s 100% transmit power contrast burst sequence followed by another 26 s of imaging (Fig. 1a). The wash-in, wash-out and reperfusion signal kinetics during the imaging sequence were monitored. The region of interest encompassing the whole tumor was drawn for each animal to generate a time–intensity curve. Percentage of contrast signal retention measured during wash-out (SR_{wo}) and on reperfusion (SR_{rp}) are defined by the equations

$$SR_{wo} = \left(\frac{C_d}{C_p} \right) \times 100\%$$

$$SR_{rp} = \left(\frac{C_r}{C_p} \right) \times 100\%$$

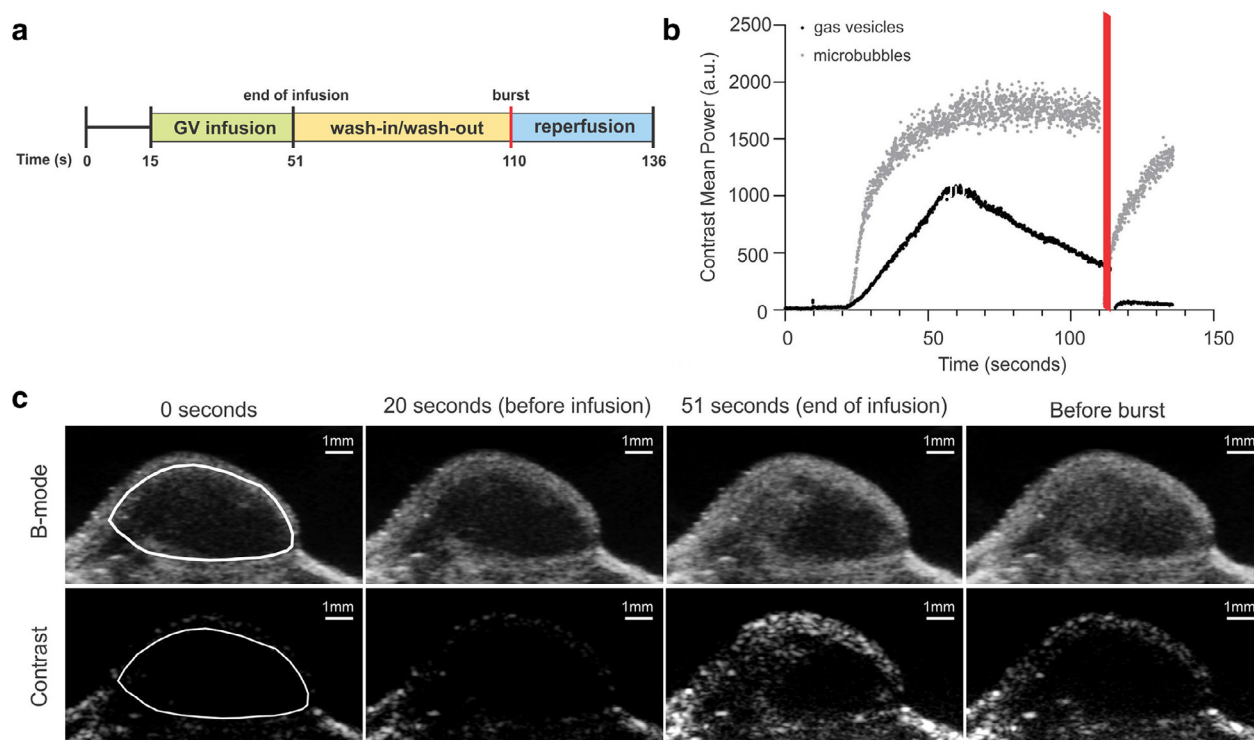


Fig. 1. Non-linear contrast signal enhancement visible in tumors after an infusion of GVs. (a) Schematic representation of imaging sequence, including time and duration of infusion, wash-in/wash-out and burst sequence. (b) Time–intensity curve for Vevo MicroMarker microbubbles and gas vesicles, with an increase in signal during GV infusion and a rapid loss of signal before a burst sequence (red line). (c) Typical B-mode and non-linear contrast mode images of subcutaneous xenograft tumors taken at indicated time points. Region of interest delineated by white line. GV = gas vesicles.

where c_p is the contrast signal mean power where peak enhancement is observed (generally at the end of the infusion at $t = 51$ s), c_d is the contrast signal mean power before the burst sequence at $t = 110$ s and c_r is the contrast signal mean power before the end of the imaging session at $t = 136$ s.

Statistical analysis

Statistical analysis was performed using the Student *t*-test unless otherwise specified. One-way analysis of variance and the Tukey *post hoc* test were used to determine significance when examining PEGylated gas vesicles. All tests were two-tailed. A *p* value < 0.05 was considered to indicate statistical significance for all tests.

Compliance with ethical standards

All procedures involving animals were approved by the Animal Care Committee at Sunnybrook Research Institute.

RESULTS

Tumour contrast imaging with GVs

GVs have previously been reported to produce robust contrast backscattered signal in the inferior vena cava and subsequently reach a steady state in the liver

(Shapiro et al. 2014). As a result, the non-linear signal generated by Halo GVs make it a promising new agent for contrast-enhanced ultrasound imaging of tumor models. A typical bolus injection of microbubbles results in a rapid wash-in of contrast agent before reaching a plateau in signal enhancement (Supplementary Fig. S1, online only). Subsequent reperfusion of microbubbles in the tumor is clearly observed after a high-power burst sequence (Fig. 1b). Backscattered signals were detected in the subcutaneous LLC xenograft tumors shortly after the start of an infusion of GVs, with peak signal enhancement at 51 s (Fig. 1c). However, unlike microbubbles, where contrast enhancement of the tumor persists after a bolus injection, signal is rapidly lost shortly after the infusion of GVs (Fig. 1b, 1c). Additionally, after a high-power burst sequence, reperfusion of contrast is clearly and reliably visible in the tumor for microbubble contrast agents but very limited with GVs. This is consistent with the lack of reperfusion in the liver observed previously (Shapiro et al. 2014). Nonetheless we illustrate, for the first time, well-visualized and quantifiable contrast enhancement produced by GV protein nanostructures in a tumor model. The apparent question of how to optimize this wash-in begins with manipulation of the RES and surface modifications to the GVs.

Disruption of the RES reduces GV contrast signal loss in tumors

The RES plays a major role in the clearance of nanoparticles (Yu and Zheng 2015). Because the GVs are nano-sized protein structures, it is plausible that the loss of contrast signal is due to the rapid clearance of the GVs from circulation by the RES. To examine this hypothesis, we saturated the phagocytic activity of the macrophage population with an initial infusion of GVs. A second infusion resulted in significant preservation of backscattered signal (Figs. 2a and 3) until the contrast burst sequence. In addition, a significant return of the contrast signal in the tumor, indicating reperfusion of the GVs, can be seen after the second injection, albeit at a much lower intensity than the peak enhancement (Figs. 2a and 3).

Intralipid was also used to disrupt the RES uptake of GVs. Intralipid is a 20% intravenous fat emulsion, made up of predominately 20% soybean oil and used clinically as a source of calories and essential fatty acids. Fat emulsions have been reported to impair the RES clearance (Fischer *et al.* 1980; Katz *et al.* 1991; Liu *et al.* 2013). An infusion of Intralipid solution through the tail vein of mice 1 h before GV contrast imaging significantly prolonged GV contrast enhancement in the tumor (Figs. 2b and 3).

In addition to Intralipid, gadolinium is an approved paramagnetic metal ion used as a contrast agent for nuclear magnetic imaging (Weinmann *et al.* 1983; Barnhart *et al.* 1987). Additionally, reports indicate that gadolinium can block macrophage phagocytic activity and selectively eliminate the Kupffer cell population (Husztik *et al.* 1980; Hardonk *et al.* 1992). Mice were first injected intravenously through the lateral tail vein with GdCl_3 and then, after a minimum of 24 h, subsequently infused with GVs and imaged for tumor contrast enhancement. Contrast enhancement in tumours of mice treated with GdCl_3 was maintained longer than in untreated animals (Figs. 2c and 3). However, for some cases with GdCl_3 treatment, no difference was observed (Fig. 2d).

A significant reperfusion of GVs in the tumor after the contrast burst sequence was observed when the macrophage population was saturated with GVs, as well as after treatment with GdCl_3 and Intralipid (Fig. 3). Collectively, the aforementioned results indicate that by disrupting the RES with known chemicals that inhibit macrophage activities, we can improve the duration of contrast enhancement in the tumors during the wash-out phase and after reperfusion.

Functionalizing GVs with PEG improves preservation of contrast enhancement in tumors

Polyethylene glycol remains one of the most widely used molecules to functionalize nanoparticles to extend systemic circulation time (Blanco *et al.* 2015). Coating

the surface of nanoparticles with PEG minimizes aggregation and shields the particles from opsonization and phagocytosis (Suk *et al.* 2015). Additionally, different molecular weights of PEGs have been reported to increase the circulation time of particles (Mori *et al.* 1991; Dos Santos *et al.* 2007; Miteva *et al.* 2015). In this study four different PEG chain lengths were attached to the surface of the gas vesicles. The commercially purchased succinimidyl carboxymethyl ester PEG chains were mixed with gas vesicles to covalently link the PEG chains onto the protein shell of the GVs. The 2-, 5- and 10-kDa PEG chain lengths protected the GVs and maintained the contrast signal enhancement in the tumour significantly longer than without PEGylation (Figs. 2e and 3). Significant reperfusion was also seen with GVs functionalized with 5- and 10-kDa PEG chains (Fig. 3). Interestingly, despite demonstrating an improvement in contrast signal enhancement duration for GVs functionalized with 2-kDa PEG chains, there was no significant improvement in contrast signal on reperfusion of 2 kDa GVs after the contrast burst sequence compared with native GVs (Fig. 3). Additionally, 15-kDa PEG did not exhibit any significant benefits compared with native GVs (Fig. 3). Peak enhancement can be seen at 51 s with a more drastic loss in signal in native GVs compared with 2-, 5- and 10-kDa PEG GVs (Fig. 4; Supplementary Fig. S2, online only). In summary, both 5- and 10-kDa PEG functionalized onto the surface of GVs protected the GVs from the RES and improved the contrast signal on reperfusion of GVs.

DISCUSSION

GVs and their functions in micro-organisms have been investigated for decades and only recently have shifted into the field of imaging research. This study is the first to use GVs as an ultrasound contrast agent for tumor imaging. Non-linear contrast enhancement produced by GVs is detectable in tumors; however, backscattered signal is rapidly lost, in <60 s, in part because of the reticuloendothelial system. Unlike MicroMarker microbubbles, which are spherical and composed of a phospholipid shell with a median diameter of 2.3–2.9 μm , gas vesicles are on the order of 100–800 nm in length and vary in shape from “lemon”-shaped to elongated rods depending on the organism producing the vesicles. Nanoparticles in the range 100–200 nm have been reported to extravasate out of the vasculature and escape the RES, whereas nanoparticles with dimensions >150 nm have an increasing likelihood of being entrapped within the liver and spleen (Blanco *et al.* 2015). Disrupting the RES through saturation or the use of gadolinium or Intralipid greatly increased the retention of gas vesicles circulating in the tumor at 95 s after

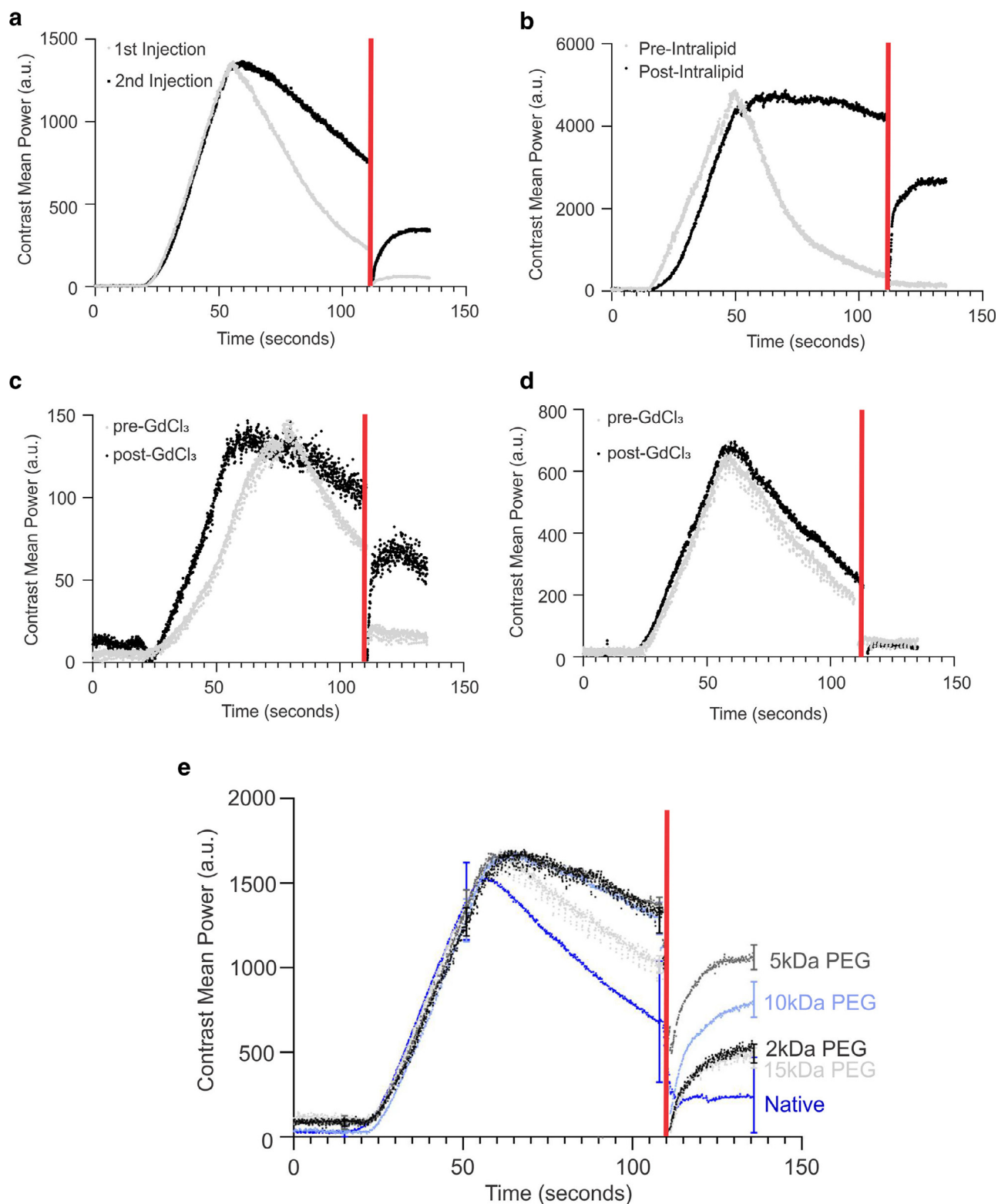


Fig. 2. Comparison of contrast circulation in tumor dynamics before and after the circumvention of the RES. Typical time-intensity curves illustrating the effect on the normalised contrast signal in tumours of (a) saturation of the RES by a dual injection of GVs, (b) treatment with 20% Intralipid, (c) treatment with GdCl₃, revealing preservation of contrast signal in the tumour and (d) treatment with GdCl₃ that did not improve retention. (e) Mean normalised time-intensity curves for native GVs (n = 21), 2-kDa PEG GVs (n = 5), 5-kDa PEG GVs (n = 7), 10-kDa PEG GVs (n = 8) and 15-kDa PEG GVs (n = 5). PEG GV standard deviations are indicated for the start of infusion (15 s), end of infusion (51 s), before burst (110 s) and at the end of imaging (136 s). GV = gas vesicles; PEG = polyethylene glycol; RES = reticuloendothelial system.

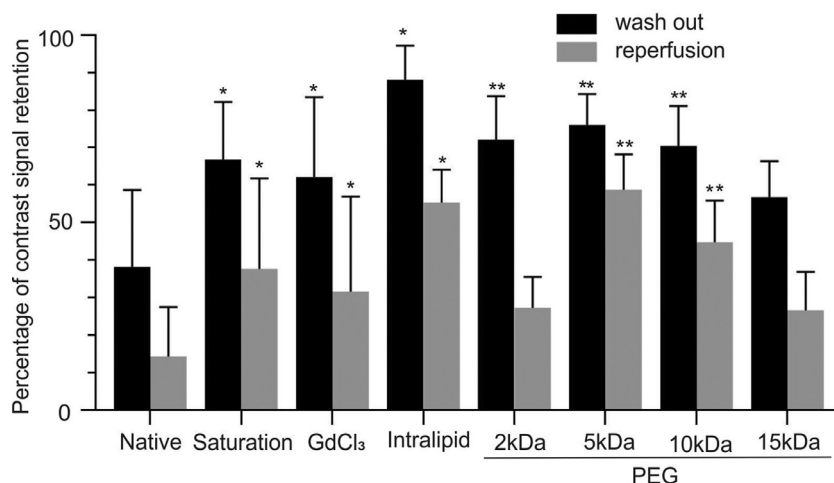


Fig. 3. Non-linear contrast signal amplitude relative to signal amplitude at peak contrast enhancement measured during wash-out just before collapse sequence at 110 s (black) and after 26 s of reperfusion (grey) for intravenously injected GVs without (Native) and with disruption of the reticuloendothelial system or modification of the GVs with PEG chains. Disruption of the Kupffer cell function by saturation ($n = 4$), treatment with GdCl_3 ($n = 6$) or treatment with 20% Intralipid ($n = 5$) resulted in a prolonged contrast signal during wash-out and improved contrast signal on reperfusion (mean $\pm$ standard deviation) compared with native GVs. * $p < 0.05$ by two-tailed Student *t*-test. Varying chain lengths of PEG bonded to the surface of the GVs affects preservation of contrast signal from GVs, with 5- and 10-kDa PEG chains resulting in increased retention of contrast signal during both wash-out and reperfusion compared with native GVs (mean $\pm$ standard deviation). ** $p < 0.05$ by one-way analysis of variance and Tukey *post hoc* test. GV = gas vesicles; PEG = polyethylene glycol.

the start of infusion compared with native GVs with contrast signals of 67%, 62% and 88%, respectively, relative to peak enhancement. After the burst sequence, the contrast signal on reperfusion was 38%, 32% and 56% of peak enhancement for saturation, GdCl_3 and Intralipid interventions, respectively, corresponding to signal increases by factors 2.7, 2.2 and 4.0 compared with native GVs. These results and the increase in contrast signal on reperfusion in particular indicate an improvement in the proportion of GVs remaining in circulation when the RES is disrupted.

Hepatocytes, also known as the parenchymal cells, constitute 60%–80% of the total liver cell population; Kupffer cells make up only 20% of non-parenchymal cells, or 2%–8% of the total liver cell population (Racanelli and Rehermann 2006). Kupffer cells are the first line of defense as they line the liver sinusoids against the endothelial cells (Bilzer *et al.* 2006). Any nanoparticles that escape Kupffer cells can extravasate out by the fenestrations in sinusoidal endothelial cells into the space of Disse and interact with hepatocytes (Racanelli and Rehermann 2006). The hepatocytes in the liver can phagocytose foreign particles and eliminate them through the hepatobiliary system (Longmire *et al.* 2012; Zhang *et al.* 2016). Evaluation of the biodistribution of radiolabelled GVs revealed a strong signal in the gall bladder suggesting that hepatocytes, as well as Kupffer cells, may be involved in the uptake of GVs (Le Floc'h *et al.* 2018) (Supplementary

Fig. S3, online only). Interestingly, treatment of mice with Intralipid resulted in a far greater improvement in GV contrast signal retention compared with gadolinium treatment, PEGylation of GVs or saturation of the liver. The result was <20% signal loss relative to peak enhancement before the burst sequence at 110 s. This can be attributed to Intralipid accumulation in hepatocytes in addition to Kupffer cells (Vilaro and Llobera 1988), thus further blocking the potential clearance of GVs from the circulation. In contrast, although GdCl_3 has been reported to inhibit activity and selectively eliminate the Kupffer cell population in the liver, residual Kupffer cells with phagocytic activity have been noted even 24 h after GdCl_3 injections in C57 BL/6 J mice (Fabre *et al.* 2010). This may explain the rapid reduction of GV contrast enhancement in the tumor observed in some animals after GdCl_3 treatment (Fig. 2d).

The shielding of GVs by coating the surface with PEG resulted in greater retention of contrast signal and improved contrast enhancement of GVs during both the wash-out phase and reperfusion. However, not all-molecular-weight PEGs tested offered the same protection from the RES, with a 2-, 5- and 10-kDa PEG chain lengths being far superior to a 15-kDa PEG chain length. GVs PEGylated with 15-kDa PEG chains did not significantly prolong the contrast signal in tumors compared with native GVs. As the PEG chains are attached to the lysine residues on the protein shell of the GVs, a 15-kDa

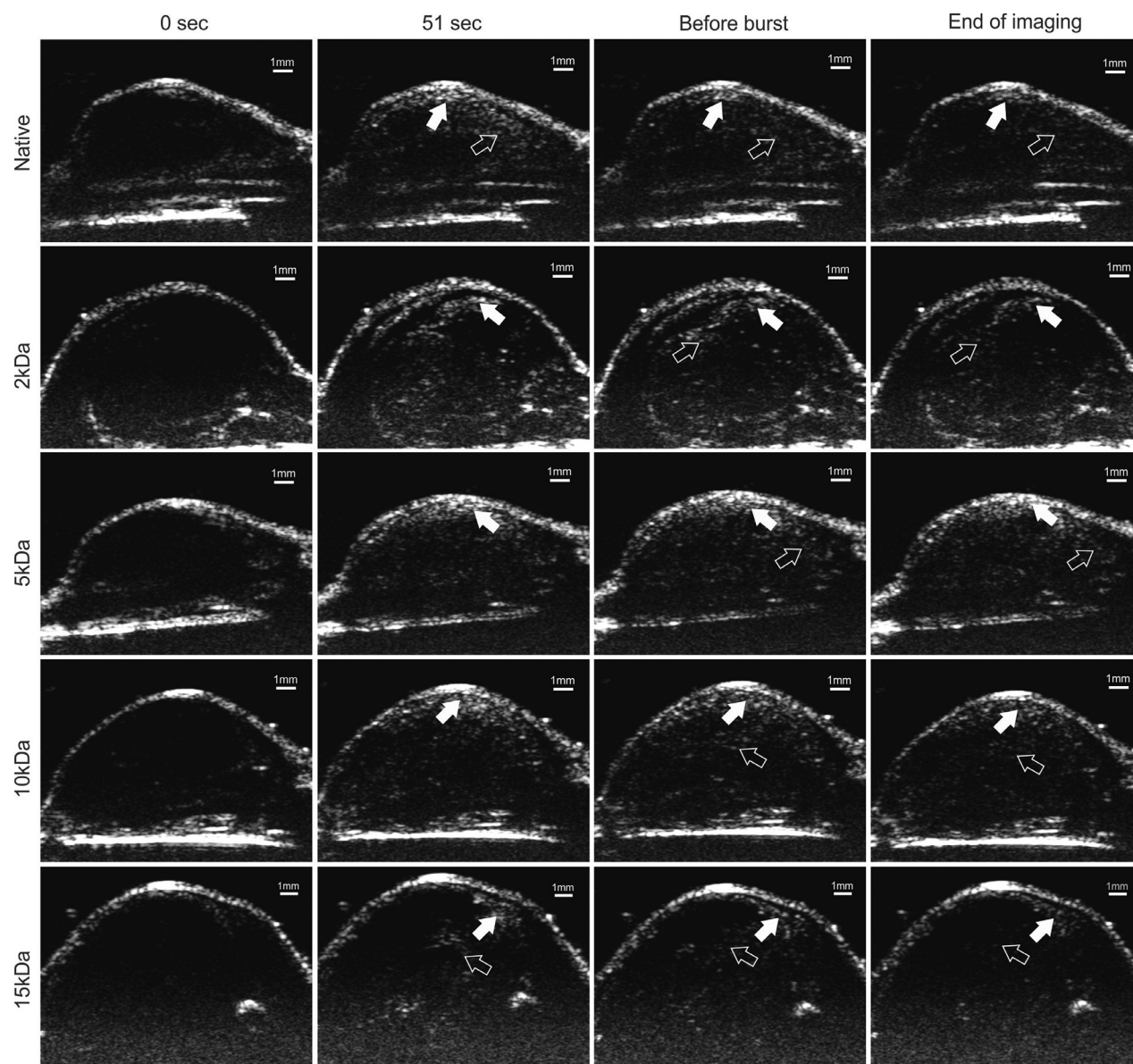


Fig. 4. Polyethylene glycol coated gas vesicles increase contrast signal retention in tumours. Typical non-linear contrast images for infusion of native GVs ($n = 21$) and GVs coated with 2-kDa ($n = 5$), 5-kDa ($n = 7$), 10-kDa ($n = 8$) and 15-kDa ($n = 5$) PEG chains at 0 s, the end of GV infusion (51 s), before a burst sequence and at the end of the imaging sequence. Regions exhibiting loss of contrast signal (*hollow arrow*) and retention of contrast signal (*solid arrow*) are evident before a burst sequence. GV = gas vesicles; PEG = polyethylene glycol.

PEG chain will more likely result in PEG–PEG steric interaction and block access to additional lysine residues nearby, thereby decreasing the overall saturation of the GV protein coat with PEG chains. GVs PEGylated with 5- and 10-kDa chains resulted in improved contrast enhancement in the tumor on reperfusion, indicating reduced clearance of GVs in tumors. Interestingly, although 2-kDa PEG improved enhancement time (72% of contrast signal preserved or 34% higher compared with native) similar to 5- and 10-kDa PEG lengths (76%

and 70% signal preservation or 38% and 32% higher compared with native, respectively), improved reperfusion was not seen with 2-kDa PEG (Fig. 3). The contrast signal remaining following reperfusion after a burst sequence was 59% and 45% preserved for 5 and 10 kDa, respectively, and only 27% preserved for 2 kDa. Currently, it is not known why there is a more drastic loss in contrast enhancement from GVs on reperfusion after a burst with 2-kDa PEGylated GVs despite having a prolonged contrast signal enhancement in the tumor during

wash-out. One possible reason could be various-molecular-weight PEG chains can alter the stability of GVs. Thus, future investigation into GV acoustic response and collapse pressure after functionalization with PEG chains is needed. Various PEG chains have been reported to affect circulation time, which may be related to physiochemical differences of the particle. Smaller PEGylated nanoparticles reduced macrophage uptake *in vitro* (Suk *et al.* 2015). Moreover, PEGylated rod-shaped nanoparticles exhibited reduced uptake by the RES compared with identical spherical-shaped nanoparticles (Arnida *et al.* 2011). Thus, in addition to PEGylation, changes to GV morphology and size may greatly influence GV circulation.

Disruption of the RES is not a feasible long-term solution to prolonging the circulation of GVs. PEGylation of GVs with 5- and 10-kDa PEG chains does improve the preservation of the contrast enhancement in the tumor during wash-out and reperfusion phase, with contrast signal relative to peak increasing by factors greater than 1.8 and 3.2, respectively, compared with non-PEGylated GVs, indicating a longer circulation in tumours. However, a substantially stronger GV signal throughout the wash-out phase and reperfusion is still needed to increase the likelihood of extravasation of GVs into target tissues and for the GVs to be used as possible extravascular molecular targeting agent. In previous work, targeted microbubbles were allowed to circulate for 4 min to allow sufficient targeting of intravascular proteins (Denbeigh *et al.* 2014; Zlitni *et al.* 2017). As a result, the circulation time of GVs must increase by at least 4 min to improve the likelihood of their use as a new targeting ultrasound agent. Recently, the protein shell of *Ana flos-aquae* GVs was genetically engineered to change the mechanical and acoustic properties of these vesicles (Lakshmanan *et al.* 2016). This carries the potential to genetically engineer the outer shell of gas vesicles to make them less likely to be detected by the RES and also provides opportunities for surface modifications that would allow GVs to act as a molecular targeting ultrasound agent. The use of GVs as a pre-clinical ultrasound agent will aid in examining pre-clinical models of disease progression, the effectiveness of intervention and the extravasation of drugs to target tissues.

CONCLUSIONS

We report, for the first time, the use of gas vesicles as an ultrasound contrast agent for potential tumor imaging. The RES plays a major role in the clearance of nanoparticles in the circulation. As a result, GVs injected intravenously are rapidly taken up by the liver, resulting in a short duration of contrast enhancement in tumors. Evading the RES by chemically blocking macrophage activity with GdCl₃ or Intralipid resulted in improved

retention of contrast enhancement from gas vesicles circulating in tumors. In addition, functionalizing GVs with 2-, 5- or 10-kDa PEG chains resulted in preservation of >70% contrast signal in the tumors. We report significant improvement in maintaining contrast enhancement from GVs in tumors by manipulating the reticuloendothelial system.

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Conflict of interest disclosure—FSF is a consultant to and receives research funding from FUJIFILM VisualSonics Inc.

SUPPLEMENTARY MATERIALS

Supplementary data related to this article can be found online at doi:[10.1016/j.ultrasmedbio.2019.09.009](https://doi.org/10.1016/j.ultrasmedbio.2019.09.009).

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